

The effects of 2D and 3D building morphology on urban environments: A multi-scale analysis in the Beijing metropolitan region

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ABSTRACT

Building morphology has critical impacts on urban environments. However, the association between building morphology and urban environments across different scales and seasons remains understudied. This study investigated the effect of 2D/3D building morphology on seasonal urban climate and air quality in the residential areas (with radii of ≤ 200 m), blocks ($= 500$ m), and communities ($= 1000$ m) around the observation stations in Beijing, using correlation and regression analysis. Results showed that the associations between building morphology and urban environments were highly scale- and season-dependent, with the 3D indicators showing stronger associations with urban environments than the 2D indicators. The correlation between building morphology and air temperature was significant at all seasons, and became stronger as the scales increased. Significant correlation was mainly found on 500- and 1000-m scales for wind velocity, and the degree was the weakest in summer. Relative humidity was largely affected by building morphology in autumn and winter, especially on small scales. Building volume density had the greatest impact on air temperature, and could explain more than 60% of its variance; building height density exerted the strongest influence on wind speed, and a 10% increase in it reduced wind speed by 1.8 m/s. Building height range and density were closely related to PM_{2.5} concentrations in spring and autumn due to local air pollutants emissions from buildings. To reduce building density with increased height and restrict building height range should be considered for urban planning in Beijing.

1. Introduction

Urbanization, the most evident expression of human modification of Earth's terrestrial surfaces, has been recognized to be a highly significant driver of local and regional climate change [1]. Although urban areas occupy a relatively small portion of our planet's land territory (i.e., 0.63% in 2010) [2], it is home to more than half of the world's population [3]. Understanding the role of urbanization on climate has received considerable attention in the broad scientific community [4–8]. The most widely observed manifestation of urbanization-induced climate change is the urban heat island (UHI), i.e., an urban area that is significant warmer than its surrounding rural area [9–11]. In addition to the UHI effect arising from modification of the surface energy budget, urban expansion also has profound impacts on air circulation, hydrological cycle, and the structure of planetary boundary layer. Moreover,

urban expansion and associated changes in climate metrics (e.g., air temperature, relative humidity, and wind) are closely related to the horizontal and vertical transport of air pollutants (e.g., particulate matter such as PM_{2.5}) and consequently, human health [6].

The urban underlying surface largely determines urban-rural differences in climate conditions, while intra-urban differences can be as large as urban-rural differences [5]. This mainly results from variations in: (1) urban structure (e.g., dimensions and spacing of buildings); (2) urban cover (e.g., built-up versus vegetated surfaces); (3) urban fabric (e.g., physical properties of concrete, asphalt, etc.); and (4) urban metabolism (e.g., energy and matter flow) in urban areas [12]. All these drivers associate with two-dimensional (2D) and three-dimensional (3D) urban morphology. The local climate zones (LCZs) represent one recent effort to explicitly link urban morphology with urban microclimate [13]. An array of urban landscape metrics has been used to define a site's LCZ

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class, including sky view factor, aspect ratio, building surface fraction, impervious/pervious surface fraction, building height, and terrain roughness [13,14]. It is no doubt that an in-depth understanding of the relationship between urban morphology and microclimate informs urban planning for climate change mitigation and adaptation in built environments [15].

Research into the relationship between urban morphology and microclimate has gone through three stages. Early research was focused on the relationship between urban land cover composition and land surface temperature [5,16,17]. Subsequently, a great deal of effort has been put into investigating the effect of land cover configuration on urban thermal environment [15,18,19]. These studies highlighted the importance of urban landscape configuration to the UHI intensity. In recent years, advances in remote sensing techniques (e.g., the development of Lidar techniques) largely facilitate 3D modeling of building infrastructure, and a growing number of studies have been focused on the relationship between 3D urban morphology and microclimate [20–24]. Studies of this kind give us new insight into how urban structure, as proposed in Ref. [12], could affect urban microclimate. For example, Ref. [23] reported that the 3D building morphology indicators outperformed the 2D indicators in predicting intra-urban variability of air temperature in Beijing. Meanwhile, the effect of urban morphology on air pollution has also been increasingly investigated [25–28].

Overall, previous research was mainly focused on urban temperature in summer on a singular scale (e.g., a 500 m-radius area) [23]. A comprehensive assessment of the relationship between building morphology and urban environments on multiple scales and how the relationship varies in different seasons remains understudied. Therefore, this study was designed to enhance understanding of the effect of 2D/3D building morphology on urban meteorology and air quality in spring, summer, fall, and winter for urban residential areas, blocks, and communities. The study area was focused on the highly urbanized Beijing, the capital city of China. The 2D/3D building morphology indicators

were extracted from the sample sites with radii of 100, 200, 500, and 1000 m around the observation stations, where air temperature, relative humidity, wind velocity, and fine particulate matter (PM_{2.5}) were recorded. Our specific research objectives were to: 1) analyze the spatial variability of 2D/3D building morphology on various scales in central Beijing; 2) investigate the relationship between building morphology and urban environments across different scales and seasons; and 3) explore the scale- and season-dependent characteristics of the relationship. Finally, we gave advice on urban planning in Beijing for better environments.

2. Study area

Beijing, the capital city of China, is located in the North China Plain, covering an area of $1.641 \times 10^4 \text{ km}^2$ and supporting a population of 21.54 million (Fig. 1). Beijing is one of the most urbanized and industrialized regions in mainland China, and urban fraction within its sixth-ring road has exceeded 75% [29]. It lies in the marginal areas of the monsoonal footprint and is characterized by a warm temperate climate. The intra-annual temperature ranges between -17.0 – 41.9°C and precipitation between 266.9–813.2 mm (<http://bj.cma.gov.cn/>). Summer (June, July, and August) in Beijing is hot and rainy, while winter (December, January, and February) is cold and dry. Spring (March, April, and May) is windy because diurnal and daily temperature is changeable in this period of the year. Autumn (September, October, and November) is sunny and rainless. Since the reform and opening-up policy in 1978, Beijing has experienced rapid urban expansion. The built-up areas of Beijing increased from $0.340 \times 10^3 \text{ km}^2$ in 1978 to $1.485 \times 10^3 \text{ km}^2$ in 2018 (nearly 5 times larger than in 1978), and ranked at the top of all Chinese cities (<http://www.mohurd.gov.cn/>). This study was focused on the central area of Beijing (i.e., the area within the 6th-ring road), where the density of buildings and population is extremely high. The central part accounts for 13.8% of the Beijing's

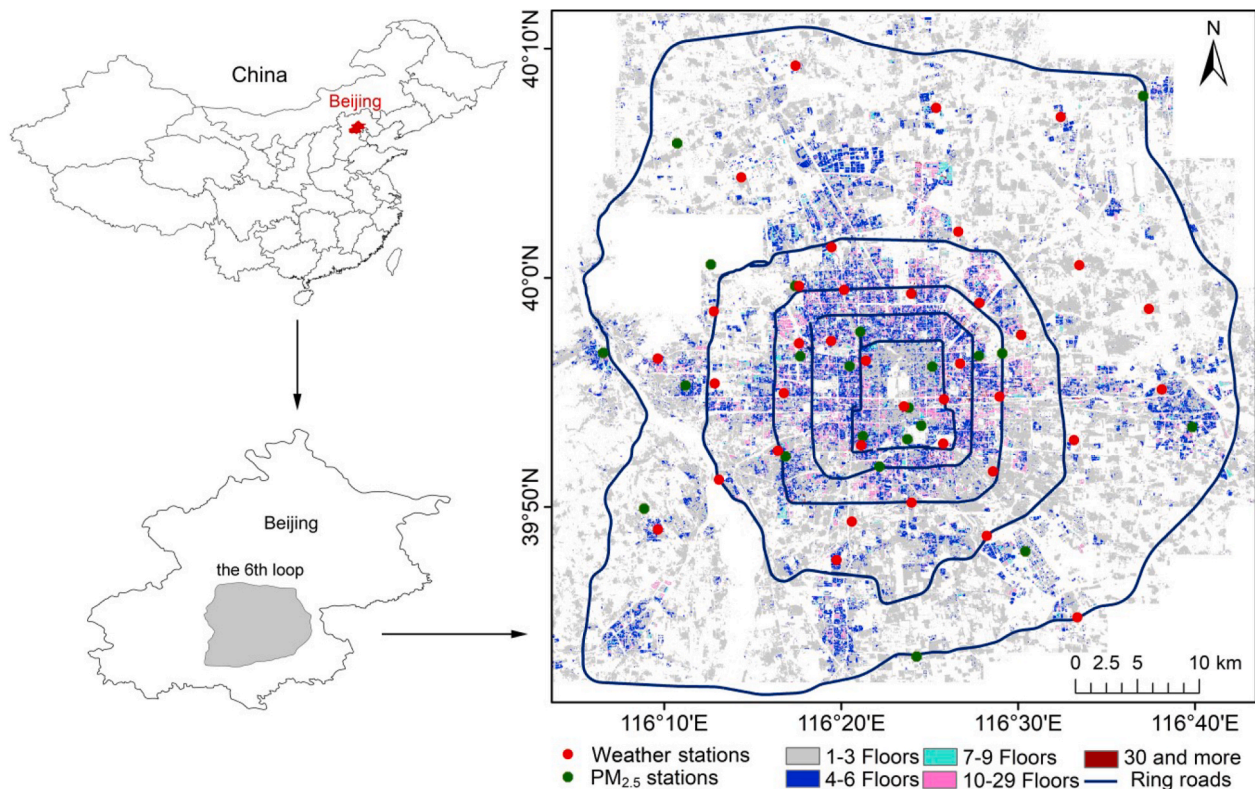


Fig. 1. On the left: the location of Beijing in China and the 6th-loop zone in Beijing. On the right: the distribution of buildings categorized by the number of floors in central Beijing, with the 37 weather stations and 24 air quality monitoring stations overlaid.

total area, with the altitude ranging between 20 and 60 m.

3. Materials and methods

3.1. Data acquisition and processing

The data used in this work included: (1) Beijing 1:2000 digital topographic map; (2) meteorological data (i.e., air temperature, relative humidity, and wind speed) from automatic weather stations; and (3) PM_{2.5} concentrations from air quality monitoring stations. A detailed description of these three kinds of data was as follows:

- The 1:2000 digital topographic map covering the urban core of Beijing was produced in 2011 by Beijing Institute of Survey and Mapping (<http://www.bism.cn/>), with a horizontal positional accuracy of 0.2 m. This large-scale digital map contained detailed information about building footprints and floor number (Fig. 1). According to the Chinese construction standard, we set the height of one floor to 3 m so that the height of buildings can be accurately estimated [24,30]. Based on the map and building height attributes, the 2D/3D building morphology indicators in specific buffer zones around the observational and monitoring stations were calculated. Generally, building density in the 4th-ring road was the highest, with low-rise buildings in the 2nd loop and high-rise buildings in between the 2nd and 4th loops. Building density between the 4th- and 5th-ring roads was relatively lower, with high-rise buildings found in the northern part. The broad area between the 5th- and 6th-ring roads was characterized by low-rise, low-density buildings.
- The hourly meteorological data were obtained from 37 ground automatic weather stations in the study area (Fig. 1). These stations were operated by Beijing Meteorological Service for weather observation and forecast (<http://bj.cma.gov.cn/>). All the stations were deployed according to national measurement standards, and the observational data must go through quality control prior to the official release. To be consistent with the date of digital topographic map, we used the observational data from January 1, 2010 at 00:00 UTC to December 31, 2012 at the same time. The 1-h data were first averaged on a daily basis, based on which the monthly and seasonal mean for a particular year was calculated. Finally, the seasonal mean was averaged across all three years (i.e., 2010, 2011, and 2012) to represent the meteorological condition of a site.
- The hourly PM_{2.5} concentrations were obtained from 24 ground monitoring stations in the study area (Fig. 1). The data were first released in 2013 by Beijing Municipal Environmental Monitoring Center (<http://www.bjmemc.com.cn/>). We used the PM_{2.5} concentrations observed from January 1, 2013 at 00:00 UTC to December 31, 2013 at the same time. Due to instrumental and anthropogenic causes, there existed missing values. To address this issue, only the stations with at least 20 records per day were selected to calculate daily average, and only the stations with at least 27 records per month (25 records in February) were used to calculate monthly and seasonal average, according to the Chinese technical regulation on ambient air quality index.

3.2. Calculation of 2D/3D building morphology

Numerous metrics have been used to quantify landscape patterns, with the 2D metrics being the most widely applied [31]. Ref. [32] calculated 55 landscape metrics for 85 land use and land cover maps, and suggested 26 metrics that could fairly represent the characteristics of a landscape. Some of these 2D metrics have been shown to be closely related to urban climate and air quality [2,18,19,26]. Based on that, we selected 6 metrics to measure the 2D building morphology in the urban core of Beijing, including the percentage of patch (PLAND), patch area range (PAR), coefficient of variation (CV), patch density (PD), largest patch index (LPI), and aggregation index (AI), all of which were

calculated for building patches. Overall, building coverage (i.e., PLAND) and the variance of building patch areas (i.e., PAR) had a direct impact on the UHI intensity and local winds [33,34]. The degree of fragmentation (i.e., CV, PD, and LPI) and aggregation (i.e., AI) of buildings associated closely with the heat transfer, air circulation, and consequently the diffusion of pollutants in urban environments [2]. The 2D metrics were calculated using FRAGSTATS 4.2 [35], with detailed calculation formulas listed in Table A1.

For the 3D metrics, existing research has shown that building height, building volume density, sky view factor, and roughness length exerted a strong influence on urban environments [34,36]. Thus, we chose the building volume density (BVD), building height range (BHR), building otherness (BO), building height density (BHD), highest building index (HBI), and sky view factor (SVF) to represent the 3D building morphology in central Beijing. The six 3D metrics were also corresponding to the 2D metrics (i.e., to use building height instead of building patch areas in the formulas), so as to explore the relative importance of the 3D and 2D metrics in regulating urban climate and air quality. A detailed description of the 3D metrics is provided in Table A2. SVF measured the degree to which the sky was obscured by the surroundings at a given point, and estimation of SVF was assumed to be the sum of the angle elements that reach the visible sky in the entire hemisphere [37]. Here, we used the method proposed by Ref. [38] to calculate SVF. As shown in Fig. 2, SVF of roads (i.e., 0.6–0.9) was typically larger than that of built-up areas (i.e., 0.1–0.5). The lowest SVF was found in the 2nd loop, generally less than 0.3.

Among these 3D building morphology indicators, SVF had the potential to regulate urban climate and air quality through incoming solar energy and ventilation in the urban canopy layer [22]. Building height (i.e., BHD and BHR), another widely used building morphology indicator, could be useful to describe the thermal properties of buildings (e.g., anthropogenic heating) [39]. It also exerted a strong influence on wind fields in urban canopy [40]. BVD was an important indicator representing the density of buildings, and had substantial impacts on the heat capacity (i.e., thermal load) of urban environments, generally through the absorption of solar energy and release of long-wave radiation back into space [6]. The roughness length of urban underlying surfaces, denoted by BO and BHR in this study, was closely related to turbulent exchange in the urban canopy layer, and had a direct impact on winds and dispersion of air pollution [41,42]. Finally, HBI was a useful indicator that reflected the height characteristics and crowding of buildings.

3.3. Creation of circular buffer zones

The 2D and 3D building morphology indicators were extracted from circular buffer zones with radii of 100 m, 200 m, 500 m, and 1000 m around the weather and air quality monitoring stations (Fig. 3a). For the Beijing metropolitan region, a buffer zone with a radius of less than 200 m corresponded to a residential area, a buffer zone with a radius of 500 m enclosed an urban block that was crossed by secondary main roads, and a buffer zone with a radius of 1000 m encircled a community that was crossed by primary main roads. As there were 37 weather stations and 24 air quality monitoring stations inside the 6th loop of Beijing, 148 (= 37 × 4) buffer zones were created and used to quantify the relationship between building morphology and microclimate at different scales, and 96 (= 24 × 4) buffer zones were created and used to quantify the relationship between building morphology and PM_{2.5} concentrations. We selected 11 samples with a radius of 1000 m to illustrate urban landscape patterns in different buffer zones (Fig. 3b). It was shown that the composition and configuration of building patches (e.g., the proportion, size, density, and degree of aggregation) varied considerably between different sample sites. Additionally, the proportion of vegetation, water, impervious surfaces, and roads also differed from one site to another.

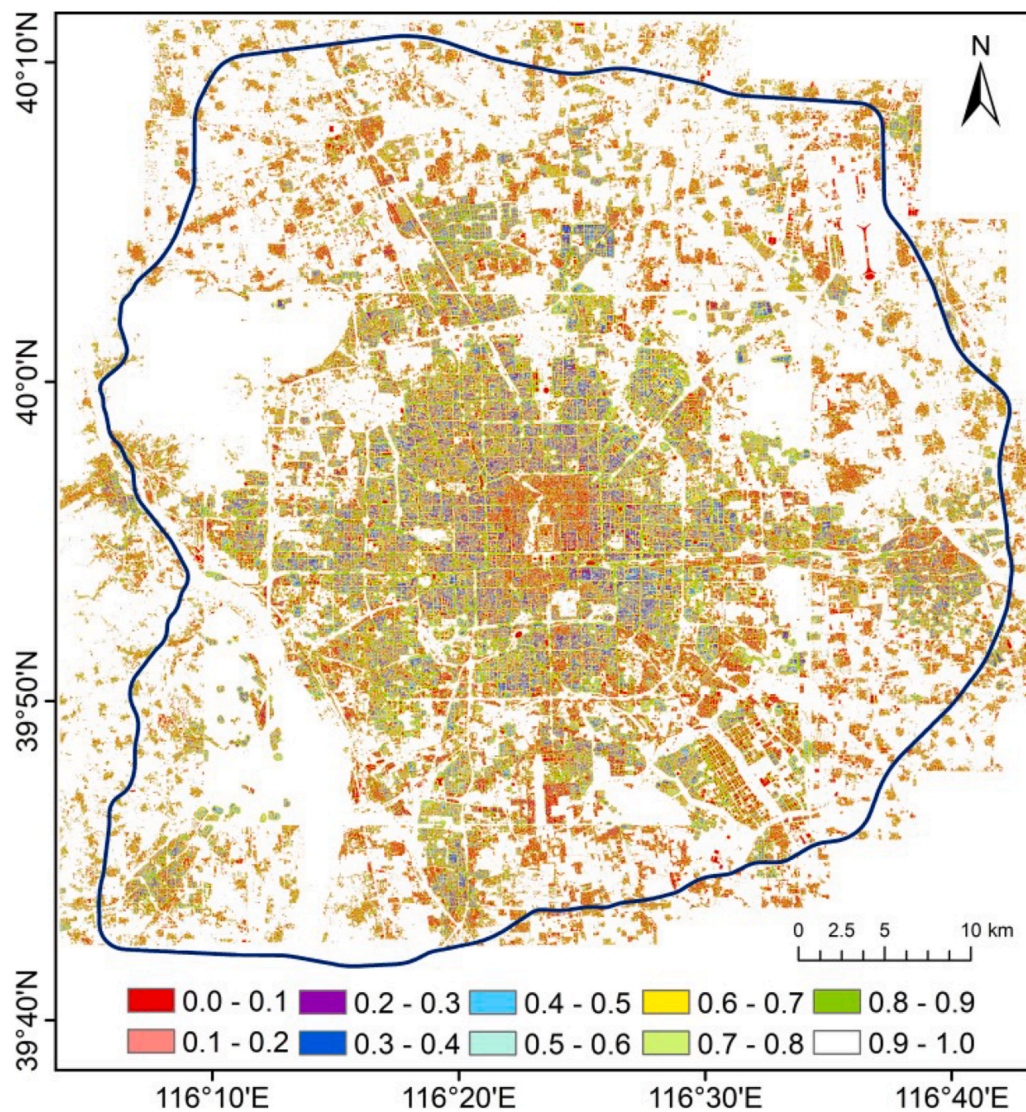


Fig. 2. The sky view factor in the 6th-loop zone of Beijing.

3.4. Correlation and regression analysis

We conducted Pearson correlation analysis to examine the relationship between building morphology and urban environments. Pearson's correlation coefficients (R) range from -1 to $+1$ and measure the strength of the linear association between two variables. Concretely, R equal to 1 indicates an ideal linear correlation with a positive slope, 0 no linear relationship between the two variables, and -1 a perfect linear correlation with a negative slope. We did the student's t -test to determine whether the correlation coefficients were significantly different from zero or not. If the p -value of the statistical test was less than 0.05, we deemed that the linear association between two variables was significant. The strength of the correlation was divided into five categories: extremely strong correlation ($0.8 \leq |R| \leq 1.0$), strong correlation ($0.6 \leq |R| < 0.8$), moderate correlation ($0.4 \leq |R| < 0.6$), week correlation ($0.2 \leq |R| < 0.4$), and extremely weak or no correlation ($0.0 \leq |R| < 0.2$).

We further carried out regression analysis for those indicators showing the strongest correlation with urban microclimate (i.e., air temperature, relative humidity, and wind velocity). The simple linear regression model was first established, using the least squares fitting technique, to investigate the explanation power (R^2) of those indicators and the regression coefficients (*slopes*) of the established equations on different scales. In addition to the univariate analysis, we subsequently

conducted stepwise multiple linear regression analysis to compare the relative importance of those indicators across various scales in affecting urban microclimate in the Beijing metropolitan region. The best fitting models were determined by a combination of AIC (Akaike's Information Criterion) and adjusted R-squared [23]. We also used AIC and adjusted R^2 for the comparison of models built for different urban climate factors.

4. Results

4.1. Urban microclimate and air quality

We first show intra-urban variations in microclimate within the 6th-ring road of Beijing using annual mean air temperature, relative humidity, and wind speed, averaged across 2010–2012, obtained from the 37 meteorological stations. Notably, urban microclimate exhibited large spatial variability, and this variability was closely related to building density. The surface air temperature in the 4th loop was clearly higher than that in between the 4th and 6th loops, with intra-urban temperature difference up to 1.88°C (Fig. 4a). On the contrary, the relative humidity in the 4th-ring road was obviously lower than that in between the 4th- and 6th-ring roads, with the intra-urban difference equal to 12.77% (Fig. 4b). Similarly, the observed wind speed at the stations located in between the 4th and 6th loops, on the order of 1.5–2.5 m/s in

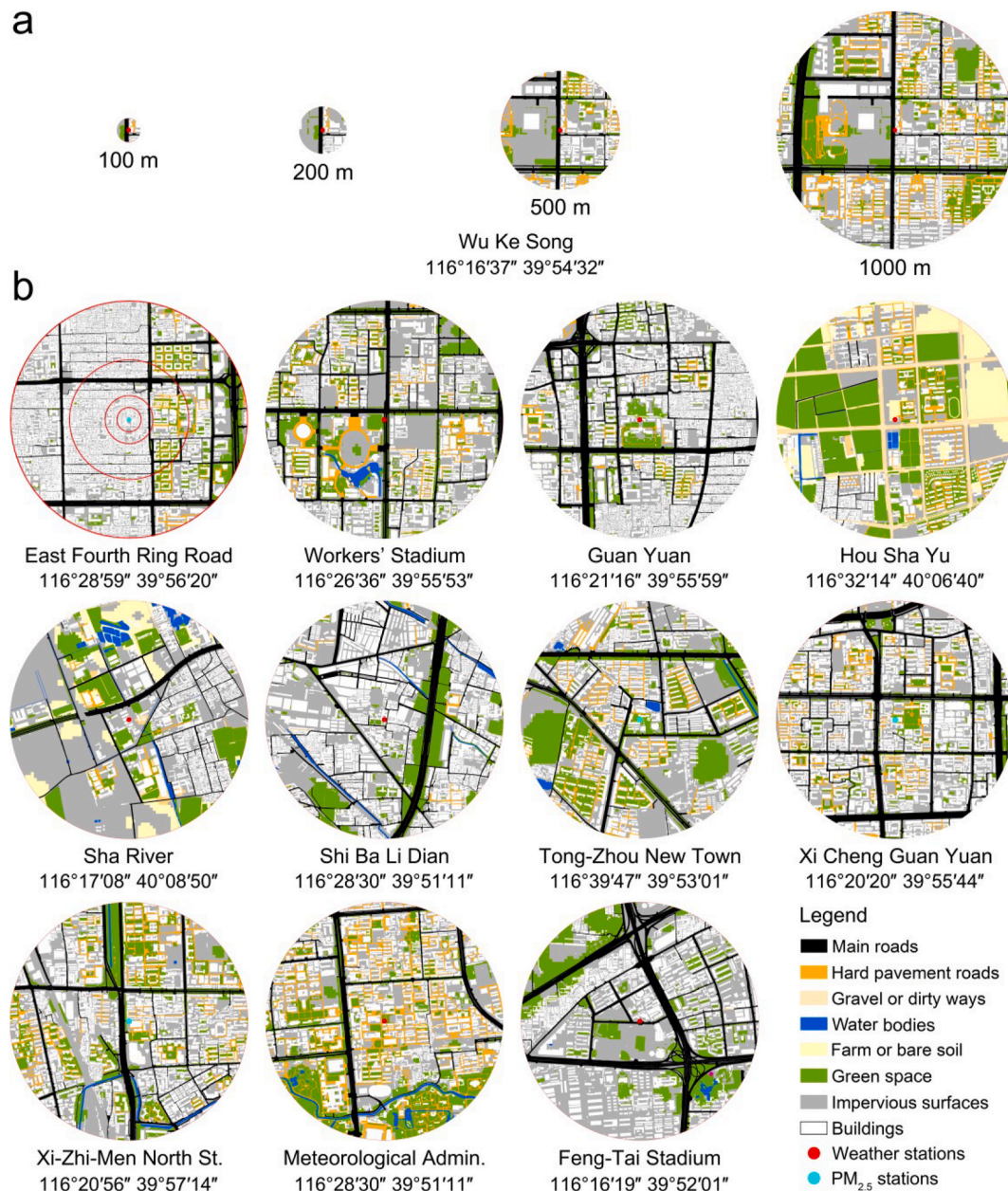


Fig. 3. Illustration of circular buffer zones with radii of 100 m, 200 m, 500 m, and 1000 m around the Wukesong weather station (a), and land cover maps in sample sites with a radius of 1000 m around the weather and air quality monitoring stations (b).

general, was considerably higher than in the core area (Fig. 4c). Inside the 4th-ring road, 9 out of 12 stations had the wind speed less than 1.2 m/s. The annual mean PM_{2.5} concentrations in the core area, as high as 142.95 $\mu\text{g}/\text{m}^3$, was generally larger than those outside the 4th loop (Fig. 4d). Exceptions were found at the three stations in the southeastern part of the area between the 5th and 6th loops, where high PM_{2.5} concentrations, with the magnitude of 100–105 $\mu\text{g}/\text{m}^3$, were also monitored. This was partly because northwesterly winds prevailed Beijing during winter, when the heaviest air pollution usually occurred.

4.2. 2D and 3D building morphology

We next show the spatial variability of building morphology on different scales at the 37 sample sites around the weather stations in the 6th-loop zone of Beijing. For the 2D indicators, PLAND within the 4th-ring road was generally larger than in between the 4th- and 6th-ring roads, and the scale effect of this indicator was inconspicuous

(Fig. 5a). Maximum PLAND was found on the 100-m scale (i.e., 47.7%), while PLAND at sample sites near the 5th- and 6th-ring roads had smaller values (i.e., <10.7%). PAR increased with the increase of scales, and PAR on the 1000-m scale was considerably larger than on smaller scales (Fig. 5b). Likewise, CV extracted from buffer zones with a radius of 1000 m was much higher than with a radius of ≤ 500 m (Fig. 5c). Except the sample sites around the ring roads, PD on 100- and 200-m scales was usually larger than on 500- and 1000-m scales (Fig. 5d). Moreover, PD on a specific scale exhibited greater spatial variability than PAR and CV. LPI within the 4th loop of Beijing was also higher than in between the 4th and 6th loops, and LPI was reduced with the increase of scales (Fig. 5e). AI ranged between 60 and 80 in general across the four scales (Fig. 5f). However, the spatial variability of AI was fairly large, especially on small scales (i.e., ΔAI decreasing from 41.9 on the 100-m scale to 13.6 on the 1000-m scale).

The spatial variability of the 3D indicators on different scales had similarity with the 2D indicators. For example, BVD inside the 4th-loop

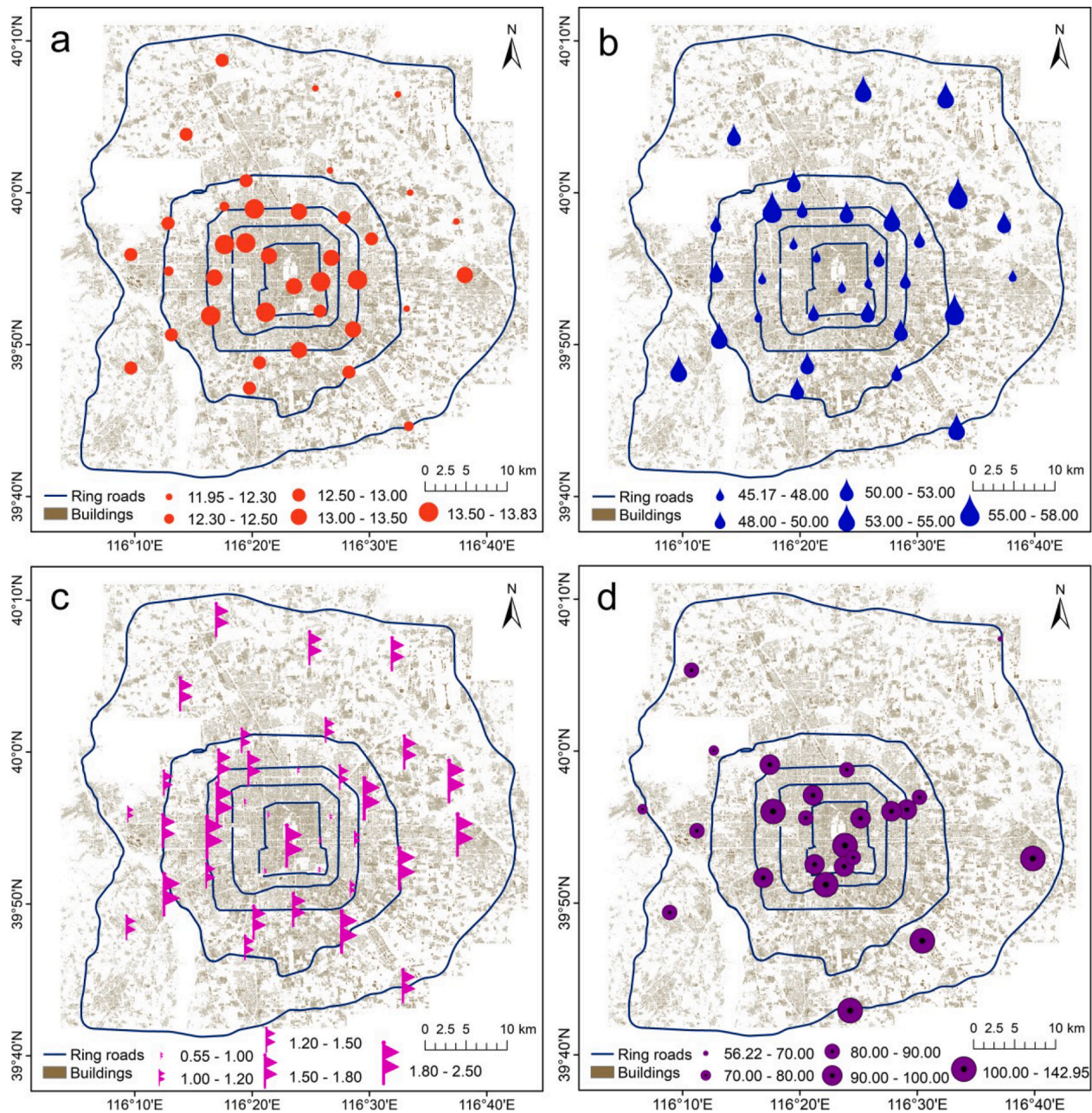


Fig. 4. Annual mean air temperature (a; °C), relative humidity (b; %), wind speed (c; m/s), and PM_{2.5} concentrations (d; µg/m³) at the 37 weather stations and 24 air quality monitoring stations within the 6th-ring road of Beijing.

zone was considerably larger than outside (Fig. 6a). On a given scale, BVD at different sample sites ranged between 0.2 and 4.4. BHR increased with the increase of scales (Fig. 6b). However, the spatial variability of BHR was higher than that of PAR, especially on 100- and 200-m scales. In addition, BHR within the 4th-ring road was generally larger than that in between the 4th- and 6th-ring roads. On 500- and 1000-m scales, maximum BHR was up to 102 m and 117 m, respectively. BO in the 4th loop was smaller than that in between the 4th and 6th loops (Fig. 6c). On the 1000-m scale, the degree of variability in BO at the 37 sample sites was low and negligible. Conversely, BHD in the 4th-ring road was apparently larger (Fig. 6d). Maximum BHD (i.e., 33.4) was found on the 100-m scale in between the 3rd and 4th loops. HBI in the core area was minor and generally less than 0.5% (Fig. 6e). Greater HBI was found at the sample sites near the 5th- and 6th-ring roads, on 100- and 200-m scales. SVF within the 4th-loop zone was also smaller than the outside areas, and SVF on 100- and 200-m scales was usually higher

than on 500- and 1000-m scales (Fig. 6f). Overall, the 3D indicators exhibited significant spatial variability and well represented the building morphology in central Beijing.

4.3. Correlation analysis

The Pearson correlation analysis revealed that the relationship between the 3D building morphology indicators and urban microclimate was stronger than the 2D indicators, and the degree of correlation between building morphology and air temperature was the highest, followed by relative humidity and wind speed. For air temperature, BVD, BHR, BHD, and SVF were significantly correlated with it (mainly at the 0.01 level) across all scales and seasons, and the degree of correlation became higher as the sample sites enlarged (Table 1). On 500- and 1000-m scales, the correlation coefficients were mostly between 0.61 and 0.80, indicating a strong correlation. In addition to SVF, BO and HBI also

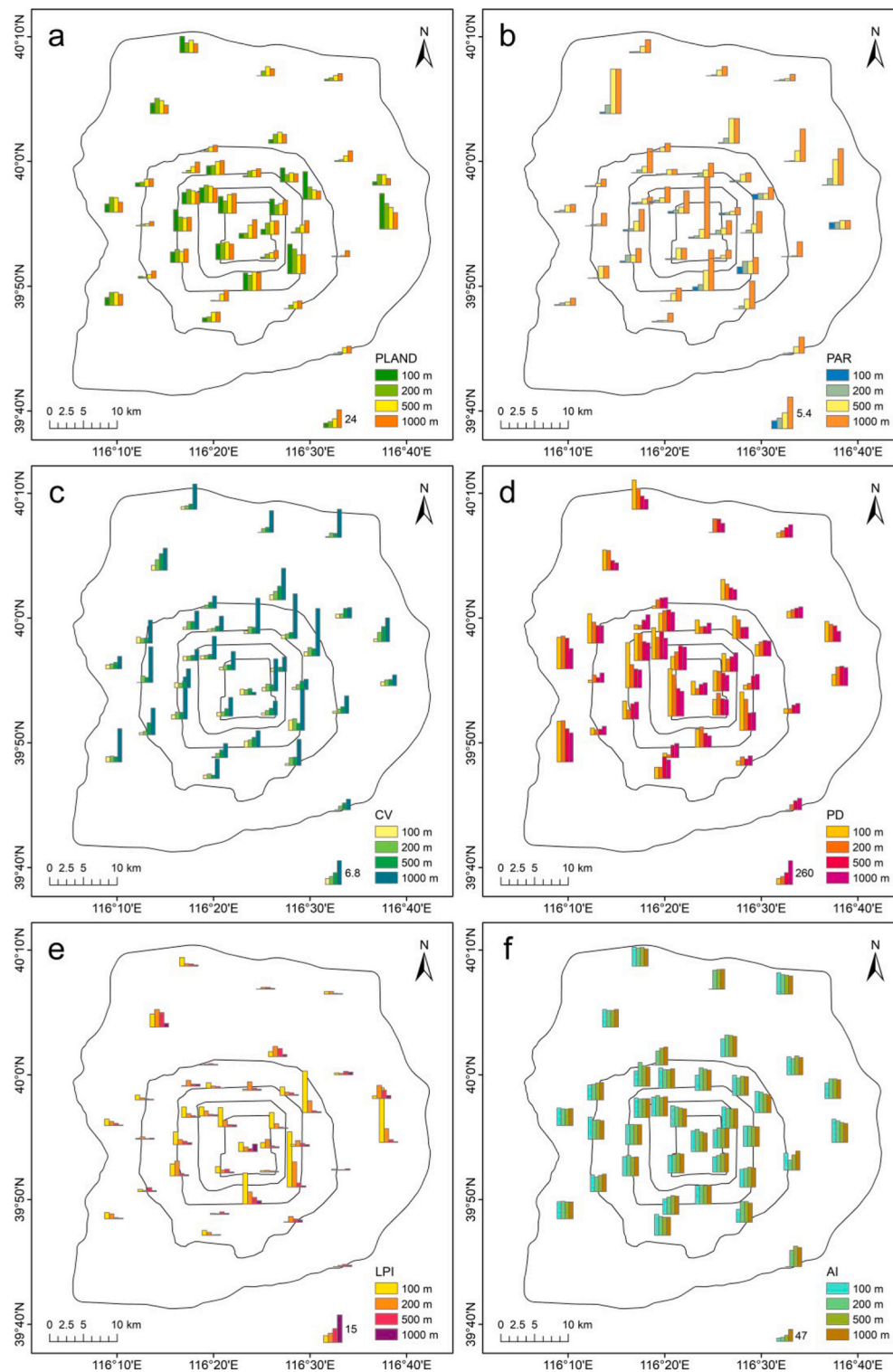


Fig. 5. The 2D building morphology indicators in sample sites with radii of 100, 200, 500, and 1000 m around the 37 weather stations in central Beijing: (a) percentage of patch (PLAND), (b) patch area range (PAR), (c) coefficient of variation (CV), (d) patch density (PD), (e) largest patch index (LPI), and (f) aggregation index (AI).

showed a negative correlation with air temperature, especially on 100- and 200-m scales. The 2D building morphology indicators showed a positive relationship with air temperature. PLAND was the only indicator significantly correlated with air temperature across all scales and seasons, with R rising from 0.35–0.51 on 100- and 200-m scales to 0.51–0.71 on 500- and 1000-m scales. PD was significantly correlated with spring, autumn, and winter temperature on 100-, 500-, and 1000-m

scales, with the strongest correlation found in winter. Finally, AI was weakly correlated with air temperature on 100- and 200-m scales through the year (i.e., $R < 0.4$).

The relationship between building morphology and relative humidity was the most complex (Table 2). For the 3D indicators, BVD, BHR, and BHD were negatively correlated with relative humidity, while BO, HBI, and SVF were positively correlated with it. BVD and SVF had a

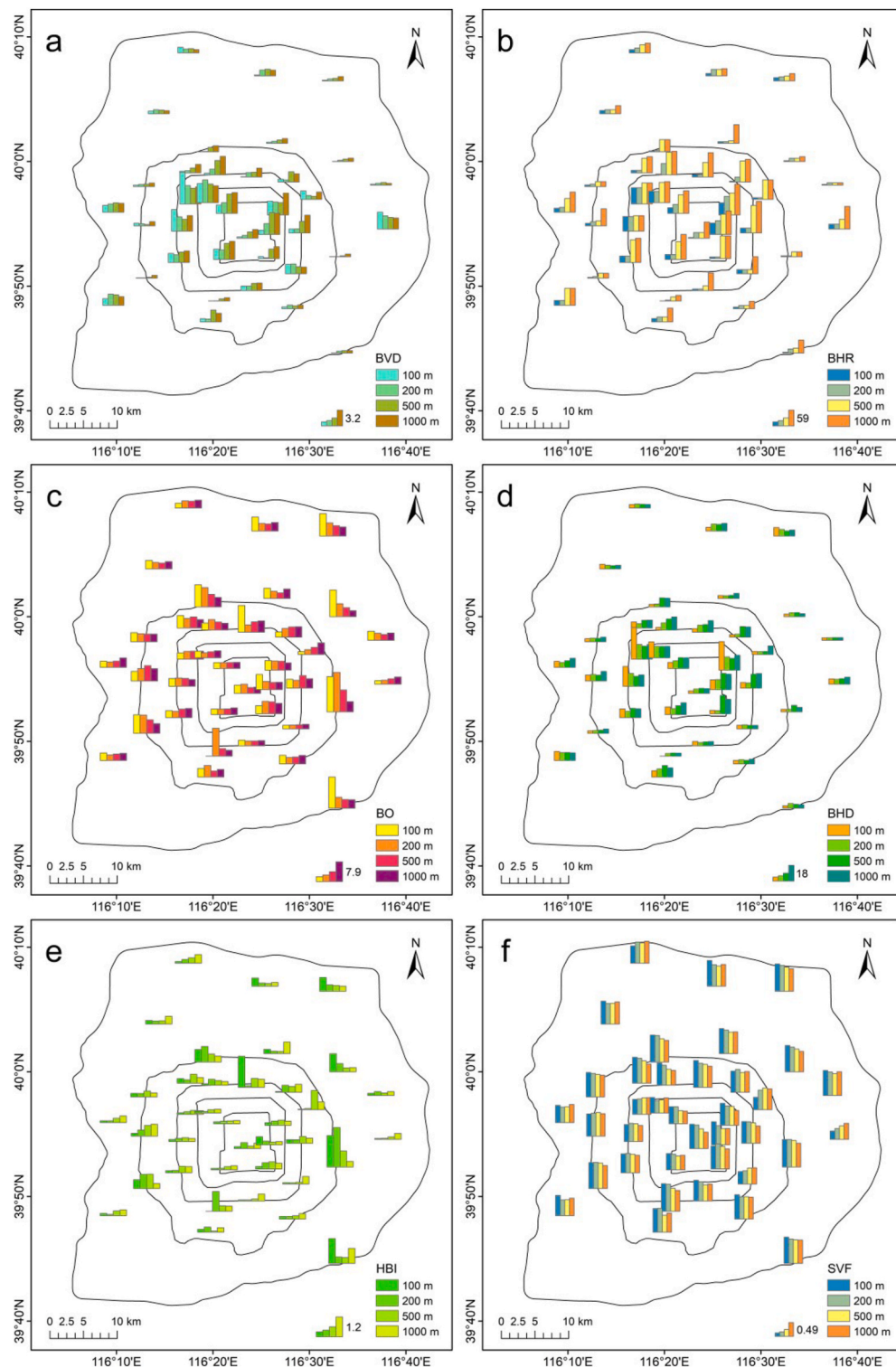


Fig. 6. The 3D building morphology indicators in sample sites with radii of 100, 200, 500, and 1000 m around the 37 weather stations in central Beijing: (a) building volume density (BVD), (b) building height range (BHR), (c) building otherness (BO), (d) building height density (BHD), (e) highest building index (HBI), and (f) sky view factor (SVF).

stronger association with relative humidity than the other indicators, with the correlation coefficients ranging between 0.51 and 0.72 in autumn and winter. Indeed, the strength of association between the 3D indicators and relative humidity was higher in autumn and winter than in spring and summer. In addition, changes in relative humidity were largely influenced by vertical building morphology within a 200-m radius around the observation stations. The 2D indicators showed a

negative correlation with relative humidity. Once again, PLAND had a significant correlation with the humidity across all scales and seasons, and the degree of correlation became higher with the increase of scales. In autumn, the correlation coefficient was up to 0.68. PD was significantly correlated with the humidity in autumn and winter on 500- and 1000-m scales. AI was greatly sensitive to scales, and was correlated with the humidity at the 0.01 level merely on the 200-m scale across the

Table 1
The correlation coefficients between the 2D/3D building morphology indicators and air temperature.

	Spring			Summer			Autumn			Winter		
	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m
<i>2D</i>												
PLAND	0.349 ^a	0.412 ^a	0.506 ^b	0.673 ^b	0.405 ^a	0.505 ^b	0.607 ^b	0.613 ^b	0.417 ^a	0.447 ^b	0.519 ^b	0.659 ^b
PD	0.349 ^a	0.473 ^b	0.372 ^a	0.473 ^b	0.347 ^a	0.440 ^b	0.607 ^b	0.613 ^b	0.342 ^a	0.426 ^b	0.462 ^b	0.575 ^b
AI	0.383 ^a	0.335 ^a							0.366 ^a	0.365 ^a	0.342 ^a	
<i>3D</i>												
BVD	0.439 ^b	0.498 ^b	0.720 ^b	0.787 ^b	0.510 ^b	0.524 ^b	0.607 ^b	0.636 ^b	0.575 ^b	0.598 ^b	0.785 ^b	0.827 ^b
BHR	0.485 ^b	0.567 ^b	0.656 ^b	0.700 ^b	0.524 ^b	0.501 ^b	0.522 ^b	0.565 ^b	0.569 ^b	0.598 ^b	0.718 ^b	0.764 ^b
BO	-0.384 ^a	-0.356 ^a			-0.472 ^b	-0.482 ^b	-0.413 ^a	-0.370 ^a	-0.366 ^a	-0.442 ^b	-0.331 ^a	
BHD	0.380 ^a	0.488 ^b	0.558 ^b	0.629 ^b	0.430 ^b	0.456 ^b	0.397 ^a	0.481 ^b	0.466 ^b	0.533 ^b	0.637 ^b	0.692 ^b
HBI		-0.351 ^a			-0.380 ^b	-0.479 ^b			-0.347 ^a	-0.365 ^a		
SVF	-0.405 ^a	-0.465 ^b	-0.613 ^b	-0.771 ^b	-0.436 ^b	-0.481 ^b	-0.562 ^b	-0.629 ^b	-0.493 ^b	-0.547 ^b	-0.659 ^b	-0.800 ^b

^a $p < 0.05$.

^b $p < 0.01$ (two-tailed).

Table 2
The correlation coefficients between the 2D/3D building morphology indicators and relative humidity.

	Spring			Summer			Autumn			Winter		
	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m
<i>2D</i>												
PLAND	-0.399 ^a	-0.452 ^b	-0.534 ^b	-0.457 ^b	-0.398 ^a	-0.435 ^a	-0.550 ^b	-0.522 ^b	-0.425 ^a	-0.421 ^a	-0.575 ^b	-0.580 ^b
PD												
AI		-0.455 ^b				-0.466 ^b			-0.373 ^a	-0.560 ^b	-0.418 ^a	-0.453 ^b
<i>3D</i>												
BVD	-0.520 ^b	-0.531 ^b	-0.478 ^b	-0.514 ^b	-0.503 ^b	-0.504 ^b	-0.498 ^b	-0.524 ^b	-0.581 ^b	-0.547 ^b	-0.655 ^b	-0.709 ^b
BHR	-0.527 ^b	-0.511 ^b		-0.364 ^a	-0.497 ^b	-0.456 ^b			-0.567 ^b	-0.549 ^b	-0.512 ^b	-0.599 ^b
BO	0.534 ^b	0.401 ^a	0.451 ^b		0.459 ^b	0.476 ^b			0.484 ^b	0.460 ^b	0.441 ^a	
BHD	-0.359 ^a	-0.456 ^b		-0.351 ^a		-0.393 ^a			-0.382 ^a	-0.537 ^b	-0.401 ^a	-0.565 ^b
HBI	0.469 ^b	0.406 ^a			0.374 ^a			0.362 ^a	0.410 ^b	0.458 ^b	0.482 ^b	
SVF	0.442 ^a	0.476 ^b	0.498 ^b	0.499 ^b	0.423 ^a	0.429 ^a	0.496 ^b	0.515 ^b	0.510 ^b	0.561 ^b	0.615 ^b	0.683 ^b

^a $p < 0.05$.

^b $p < 0.01$ (two-tailed).

four seasons.

The relationship between building morphology and wind speed was highly scale-dependent (Table 3). For the 2D indicators, only PD was significantly and negatively correlated with wind speed, especially in spring and winter on 500- and 1000-m scales. Significant correlation between the 3D indicators (i.e., BVD, BHR, BHD, and SVF) and wind speed was only found on 500- and 1000-m scales, and the strength of the relationship was weaker in summer compared with the other seasons. Among the 3D indicators, BHD had the strongest correlation with wind speed, while SVF had the weakest. Overall, the correlation coefficients between the 2D/3D building morphology indicators and wind speed ranged between 0.42 and 0.58 at the 0.01 level, indicating a moderate correlation. It was assumed that the association between building morphology and urban climate factors could influence the association between that and PM_{2.5} concentrations. However, results showed that only BHR and BHD, building height indicators which represented the thermal properties of buildings (e.g., anthropogenic heating), were significantly and positively correlated with PM_{2.5} concentrations in spring and autumn, mainly on 100- and 200-m scales (Table 4). This meant that local emissions of air pollutants from buildings might play an important role in affecting PM_{2.5} concentrations. BHD, by comparison, had a stronger correlation with PM_{2.5} concentrations than BHR, and the correlation coefficients were up to 0.68–0.69 in autumn, indicating a strong correlation.

4.4. Regression analysis

We further conducted simple linear regression analysis for the four indicators showing the strongest correlation with urban microclimate on an annual scale: BVD, BHR, BHD, and SVF (Fig. 7). In terms of air temperature, although R^2 increased as the sample sites enlarged, the slopes of regression lines did not show the same case. BVD was the strongest predictor, explaining 55.60% and 63.26% of the variance in air temperature on 500- and 1000-m scales. A 10% increase in BVD increased air temperature by 0.23 °C on both scales. SVF explained 41.12% and 61.56% of its variance, and a 10% increase in SVF resulted in near-surface cooling of 0.19 °C and 0.27 °C, respectively. Although the explanation power of BHD was comparatively weaker, the regression coefficients were the largest among the four indicators. A 10% increment in BHD could raise air temperature by 0.30 °C, on the 200-m scale. For relative humidity, BVD, once again, was the strongest predictor, explaining 43.01% of the variance in it on the 1000-m scale. A 10% increase in BVD reduced relative humidity by 0.7%–0.8%. By comparison, a 10% increase in BHR and BHD led to a 1.5% decrease in relative humidity, on 100- and 200-m scales respectively. For wind speed, R^2 was less than 30% for all the indicators. In addition, the explanation power and regression coefficients were close on 500- and 1000-m scales. A 10% increase in BVD, BHR, and SVF brought about changes in wind speed by 1.0–1.2 m/s, while for BHD, it was 1.8 m/s.

Finally, we conducted stepwise multiple linear regression analysis to investigate the combined effects of the 3D building morphology indicators on annual meteorology in the Beijing metropolitan region on 100-, 200-, 500-, and 1000-m scales, respectively (Table 5). Results showed that the explanation power of the 3D indicators to air temperature strengthened as the scales increased. On the 100-m scale, BHR and BO explained 39.4% of the variance in air temperature; on the 1000-m scale, BVD and SVF were the most important predictors, and the two indicators explained 65.0% of the variance in air temperature. On the contrary, the explanation power of the 3D indicators to relative humidity weakened as the scales increased from 100 to 500 m. BHR and BO, once again, were the strongest predictors on the 100-m scale, and explained 28.1% of the variance in relative humidity. SVF had a large impact on relative humidity on the 1000-m scale, and it alone explained 31.2% of the variance in relative humidity. However, the AIC of the regression models was more than 160.0 on all scales, indicating that the models might be overfitted. Regarding wind speed, BHD was the only

Table 3
The correlation coefficients between the 2D/3D building morphology indicators and wind speed.

	Spring				Summer				Autumn				Winter			
	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m
2D																
PD	-0.338 ^a		-0.443 ^b	-0.479 ^b									-0.356 ^a	-0.363 ^a	-0.439 ^b	-0.497 ^b
3D																
BVD			-0.533 ^b	-0.554 ^b			-0.350 ^a	-0.393 ^a				-0.383 ^a			-0.535 ^b	-0.539 ^b
BHR			-0.535 ^b	-0.526 ^b			-0.391 ^a	-0.381 ^a				-0.462 ^b			-0.525 ^b	-0.540 ^b
BHD			-0.575 ^b	-0.542 ^b			-0.443 ^b	-0.427 ^b				-0.494 ^b			-0.572 ^b	-0.528 ^b
SVF			0.348 ^a	0.441 ^b								0.336 ^a			0.373 ^a	0.458 ^b

^a $p < 0.05$.

^b $p < 0.01$ (two-tailed).

Table 4The correlation coefficients between the 2D/3D building morphology indicators and PM_{2.5} concentrations.

	Spring				Summer				Autumn				Winter			
	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m
3D																
BHR	0.423 ^a								0.622 ^b	0.458 ^a						
BHD	0.465 ^a	0.565 ^b	0.413 ^a						0.690 ^b	0.680 ^b	0.464 ^a	0.396 ^a				

^a $p < 0.05$.^b $p < 0.01$ (two-tailed).

indicator entered into the regression models, and it explained 26.7% and 23.9% of the variance in wind speed on 500- and 1000-m scales, respectively. The stepwise regression analysis also revealed that the relationship between building morphology and air temperature was the strongest, with the adjusted R^2 being the highest and AIC the lowest.

5. Discussion

5.1. Impacts of building morphology on urban environments

This study focused on the relationship between 2D/3D building morphology and urban environments across different scales and seasons in the Beijing metropolitan region. Compared with previous research examining intra-urban temperature variability on a singular scale [21, 23,43], our study showed that the association between building morphology and urban microclimate could be highly scale-dependent. The scale effect was demonstrated in the three aspects. First, some indicators were spatially homogeneous on particular scales and therefore had no significant correlation with intra-urban climate variability on those scales (e.g., AI and BO on the 1-km scale). Second, on the contrary, the correlation between the building morphology indicators and urban microclimate (e.g., wind velocity) might be significant only on particular scales. Third, the correlation coefficients became larger as the scales increased for some indicators (e.g., PLAND, BVD, and SVF). This stimulated our attention that we should choose the appropriate building morphology indicators to predict urban meteorology on a given research radius [24], and for a specific meteorological element, we should select the appropriate scales to best predict it [44]. For example, the 500–1000 m scales were suitable to predict air temperature and wind speed, while smaller scales might be suitable for relative humidity.

In addition to the scale effect, the effect of building morphology on urban environments was season-dependent. It was shown that building morphology exerted a stronger influence on relative humidity during the dry season (i.e., autumn and winter), when the moisture content in the atmosphere was low over Beijing. That is, the drier the environment was, the greater the impact of building morphology had on atmospheric moisture content. Wind speed in spring, autumn, and winter exhibited a stronger association with building morphology than in summer. Typically, Beijing was windy during spring and winter, with summer having minimum wind velocity throughout the year. On windy days, the wind-blocking effect due to the presence of buildings became evident, which had considerable impacts on the speed and direction of winds [45]. On windless days, however, such an effect was largely weakened. Compared with humidity and winds, the effect of building morphology on air temperature had no marked season-dependent characteristics, indicating that building morphology had the most direct impact on air temperature (i.e., through the absorption of solar energy and partitioning of available energy into sensible and latent heat) [7]. It should be noted that observation uncertainty existed due to the limited weather stations within the urban area, and future research may use models to simulate the effect of building morphology on urban microclimate.

The association between building morphology and PM_{2.5} concentrations was significant in spring and autumn rather than in summer and winter. Usually, Beijing had the highest PM_{2.5} concentrations in winter (e.g., 127.85 $\mu\text{g}\cdot\text{m}^{-3}$ in 2013) and the lowest in summer (e.g., 81.05

$\mu\text{g}\cdot\text{m}^{-3}$ in 2013) [46]. Winter was controlled by the high-pressure system over the capital city and was favorable for the formation of temperature inversion. Such weather conditions were not conducive to the diffusion of the PM_{2.5} pollutants, which were mainly originated from fossil fuel combustion [25,46,47]. On the contrary, summer in Beijing was dominated by the low-pressure system with strong air convection, and the composition of air pollutants was relatively simple (i.e., mainly secondary sulphur) [47]. The synoptic weather conditions and air pollution sources determined that building morphology had minimal impacts on PM_{2.5} concentrations in winter and summer. This was consistent with previous research reporting a poor fitting result of the relationship between urban landscape patterns and winter PM_{2.5} pollutions in Beijing [25]. However, air pollution sources were diverse and the weather was changeable in spring and autumn [47]. Thus, local pollutants emissions from buildings contributed largely to PM_{2.5} concentrations. Meanwhile, the associations between building morphology and urban climate factors might also exert an influence on PM_{2.5} concentrations. For example, a reduced wind speed owing to the presence of buildings might increase local PM_{2.5} concentrations.

Compared with previous research examining urban green space, roads, and water bodies, this study focused specifically on buildings. This helped understand how, and to what extent, buildings affected urban environments. Results showed that PLAND, PD, and AI had a stronger correlation with urban microclimate than the other 2D indicators, and the degree of correlation was comparable with the 3D indicators. This was generally consistent with previous research [19,26], although discrepancies existed. For example, Ref. [18] reported that LPI had the strongest association with land surface temperature in an urban area of Baltimore County, while PD was weakly correlated with land surface temperature. The discrepancies were mainly attributed to the intensity of urbanization in different part of the world. As the capital city of China, buildings in Beijing were densely distributed [7], and the size of building patches in a sample site might not have large differences. This was confirmed by the small values of PAR, CV, and LPI on given scales. As a result, these indicators had no significant correlation with urban microclimate in the Beijing metropolitan region. Although we did not examine the non-built-up patches (e.g., urban parks, vegetated road sides, and lakes), these patches might have a greater impact on the variability of climate in less developed urban areas [22].

This study also showed that the 3D indicators outperformed the 2D indicators in terms of the linkage between building morphology and urban environments. Undoubtedly, the increasing availability of lidar data and advances in motion processing techniques have facilitated research on 3D analysis in recent years [48]. Currently, the 3D metrics were rooted in traditional, patch-based landscape metrics, with building morphology characterized by footprint, height, volume, and a more comprehensive indicator SVF [34]. These metrics had complex effects on urban climate and air quality that extend beyond findings of this study. For example, previous research has shown that the effect of SVF on air temperature could be positive [49], negative [20,43,50], and neutral [51], dependent on the dominant process that drove such changes (i.e., the cooling effect of ventilation or the warming effect of enhanced energy absorption due to increased SVF) [22]. The response of daytime and nighttime temperature to building morphology may also be different. Our additional analysis showed that the association between

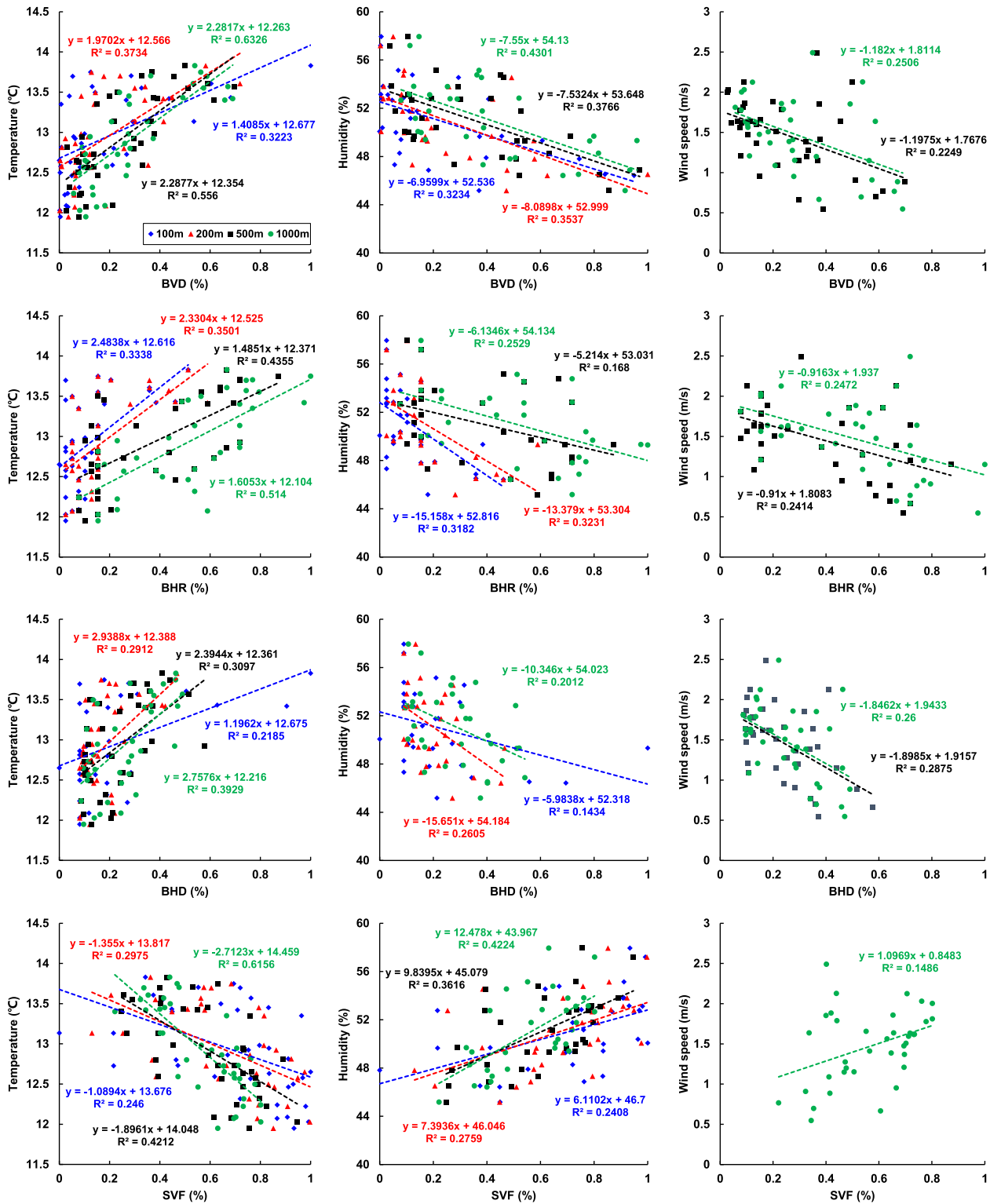


Fig. 7. The relationship between the 3D building morphology indicators (i.e., BVD, BHR, BHD, and SVF) and urban climate metrics (i.e., air temperature, relative humidity, and wind speed) on 100-m, 200-m, 500-m, and 1000-m scales.

building morphology and air temperature was stronger in the nighttime than in the daytime (Tables B1 and B2). This was mainly because: 1) buildings absorb energy in the daytime and release the energy in the nighttime, thus leading to considerable nighttime warming; 2) the shadowing effects of buildings suppress local temperature in the daytime, especially for sunny seasons [24]. Given the complex climate effects, future research should develop more physically significant 3D

metrics for urban climate research.

5.2. Implications for urban planning

This study showed that building morphology (i.e., the proportion, density, and height of buildings) exerted a strong influence on urban environments in Beijing. Due to the high heat-storage capacity and

Table 5

The standard coefficients between the 3D building morphology indicators and urban climate factors.

	Air temperature				Relative humidity				Wind speed			
	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m	100 m	200 m	500 m	1000 m
BVD		0.611 ^b	0.746 ^b	0.463 ^a			−0.464 ^b					
BHR	0.504 ^b				−0.360 ^a	−0.290						
BO	−0.316 ^a				0.369 ^a							
BHD											−0.536 ^b	−0.524 ^b
HBI						0.340						
SVF				0.384				0.578 ^b				
Adjusted R ²	0.394	0.356	0.543	0.650	0.281	0.215	0.190	0.312			0.267	0.239
AIC	45.84	47.22	34.48	25.55	166.32	169.13	169.23	164.00			37.56	38.96

^a $p < 0.05$;^b $p < 0.01$ (two-tailed).

impediments to atmosphere motion from the construction of 3D buildings, any infill development in Beijing should proceed with caution. Instead, we estimated that a 20% decrease in building volume density reduced surface air temperature by 0.46 °C, indicating that the UHI intensity could be restricted to 1.5 °C according to China's construction standard for urban residential areas (the UHI intensity was about 1.9 °C during 2010–2012 in Beijing) [24]. However, for such a densely populated region, large-scale removal of buildings may not be practical. Alternatively, to heighten the existing buildings while reducing the density of buildings should be a practical solution. The high-rise, low-density building arrangements with greater sky view factor have the potential for increasing albedo, stimulating air circulation, and generating the shadowing effects [22,24,52].

The vertical layout of buildings (i.e., roughness length) was also important. It was shown that the reduction of building height range lowered near-surface temperature, increased moisture content, and intensified wind speed. The restriction of building height differences matters because more solar energy will be absorbed as the range of building height becomes larger [52]. However, this did not mean that building height in Beijing should be homogenous. The increase of building otherness produced the same climate effects as the decrease of building height range. That is, non-uniform height had to be maintained meanwhile, and we estimated that building height range of less than 60 m might be appropriate (i.e., to ensure annual temperature < 13 °C, wind speed > 1.3 m/s, and relative humidity > 50% on the 500-m scale). Overall, the association between building morphology and urban environments was complex, and urban planners should make a comprehensive consideration and assessment of the design options.

6. Conclusions

This study established a multi-scale analysis of the relationship between building morphology and urban environments across different seasons in the Beijing metropolitan region, using correlation and regression analysis. Six 2D building morphology indicators were selected, based on which the 3D indicators were developed. The building

morphology indicators were extracted from buffer zones with radii of 100, 200, 500, and 1000 m around the weather and air quality monitoring stations. Urban climate factors (i.e., air temperature, relative humidity, and wind velocity) during 2010–2012 and air quality (i.e., PM_{2.5} concentrations) in 2013 were obtained and examined. Overall, the 3D indicators had a stronger association with urban environments than the 2D indicators. Among the 2D indicators, PLAND, PD, and AI exerted the strongest influence on urban climate, while for the 3D indicators, they were BVD, BHR, BHD, and SVF. The degree of associations between building morphology and air temperature got higher as the scales enlarged. BVD had the largest impact on air temperature, and could explain more than 60% of the variance in temperature. Wind speed was significantly correlated with building morphology mainly on 500- and 1000-m scales, and the degree of correlation was weaker in summer compared with the other seasons. BHD had the greatest influence on wind speed, and a 10% increase in it lowered wind speed by 1.8 m/s. Relative humidity was largely affected by building morphology in autumn and winter, especially on small scales (≤ 200 m). Only BHR and BHD were significantly correlated with PM_{2.5} concentrations, in spring and autumn on small scales in particular, indicating that local emissions of air pollutants from buildings might play a dominant role. To reduce building density and restrict building height range were important and should be considered for urban planning in Beijing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.buildenv.2021.107635>.

Table A1

Summary of the 2D building morphology indicators used in this study

Metrics	Abbreviation	Formula	Description
Percentage of patch	PLAND	$\frac{\sum_{i=1}^n a_i}{A} \times 100\%$	where a_i is the area of building patch i and A the buffer area
Patch area range	PAR	$a_{max} - a_{min}$	where a_{max} is the area of the largest building patch and a_{min} the area of the smallest building patch
Coefficient of variation	CV		where a_i is the area of building patch i and $\bar{a}$ the average area of building patches

(continued on next page)

Table A1 (continued)

Metrics	Abbreviation	Formula	Description
Patch density	PD	$\frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - \bar{a})^2}}{\bar{a}}$	where n is the number of building patches and A the buffer area
Largest patch index	LPI	$\frac{\frac{n}{A} \max(a_i)}{100\%}$	where $\max(a_i)$ is the area of the largest building patch and A the buffer area
Aggregation index	AI	$\frac{g_{ii}}{\max \rightarrow g_{ii}} \times 100\%$	where g_{ii} is the number of like adjacencies (joins) between pixels of buildings and $\max \rightarrow g_{ii}$ the maximum number of like adjacencies between pixels of buildings

Table A2

Summary of the 3D building morphology indicators used in this study

Metrics	Abbreviation	Formula	Description
Building volume density	BVD	$\frac{\sum_{i=1}^n V_i}{A}$	where V_i is the volume of building i (volume = footprint $\times$ height) and A the buffer area
Building height range	BHR	$H_{\max} - H_{\min}$	where $H_{\max}$ is the height of the tallest building and $H_{\min}$ the height of the lowest building
Building otherness	BO	$\frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (H_i - \bar{H})^2}}{\bar{H}}$	where H_i is the height of building i and $\bar{H}$ the average height of buildings
Building height density	BHD	$\frac{\sum_{i=1}^n H_i}{A}$	where H_i is the height of building i and A the buffer area
Highest building index	HBI	$\frac{\sum_{i=1}^n \max(H_i)}{\sum_{i=1}^n H_i} \times 100\%$	where $\max(H_i)$ is the height of the tallest building and H_i the height of building i
Sky view factor	SVF	$2\pi \cdot \left[1 - \frac{\sum_{i=1}^n \sin \gamma_i}{n} \right]$	where γ_i is the vertical angle of the horizon in the direction i

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